

1 302

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2025

Programme: BE

Full Marks : 100

Course: Cloud Computing and Virtualization

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define cloud computing according to NIST. Explain the characteristics of cloud computing. 8
- b) Describe the Jericho Cloud Cube Model and explain how it helps organizations decide cloud adoption strategy. 7
2. a) Compare SaaS, PaaS, and IaaS in terms of architecture, responsibilities, and use cases. Provide example for each service models. 8
- b) Discuss Virtual Private Network (VPN), DNS service, and cloud networking concepts. 7

OR

Explain virtual private cloud (VPC) architecture including subnets, VPN, DNS services and multihoming.

3. a) Define Hypervisor. Compare and Contrast the Types of Hypervisors based on architecture, usage and applications. 8
- b) What do you mean by virtualization in cloud computing? Explain the different levels of Virtualization. 7
4. a) Compare object storage, file storage, and block storage in terms of architecture, use cases, and performance. 7
- b) Design a cloud storage architecture using object, block, and file storage for an enterprise system. Justify your choices with redundancy and durability considerations. 8

OR

Explain Database as a Service (DBaaS). Compare SQL and NoSQL databases for cloud-based applications.

5. a) What is SLA in cloud computing? Explain different types of SLAs and their importance. 7

- b) Explain different scaling strategies in cloud computing. How do they help optimize workload distribution? 8
6. a) Discuss the role of Identity and Access Management (IAM) in cloud security. How does IAM contribute to secure resource access? 8
- b) Discuss disaster management in cloud computing and design a disaster recovery strategy for a cloud-hosted service. 7

Write short notes on: (Any two)

2×5

- a) Cloud Computing Applications
- b) Parallel File System vs Distributed File System
- c) Cloud Security Threats